ATIQUE AHMAD

Python Software Developer | AI/ML Engineer | Data Team Lead
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Summary

Python developer and AI/ML engineer with nearly 4 years of experience building production Python applications, REST APIs, data pipelines and LLM-powered systems. I design and integrate databases (PostgreSQL, MongoDB), automate data processing and reporting, and work hands-on with LLMs and embedding models for retrieval, search and verification. I have led the data team that built our in-house training datasets, and I work closely with backend, frontend and product teams to turn AI capabilities into reliable business features.

Key Achievements

- **Led the dataset team** that curated **20,000+ labelled in-house data points** across two production AI models, from collection and labelling to cleaning and training-ready splits.
- Built and shipped **Python/FastAPI services on AWS** that power the platform's AI verification and evaluation features, integrated with Go, React and LangGraph services.
- Fine-tuned a **2B-parameter model to 91.7% accuracy**, outperforming GPT-5 (+5.4 pts) on the same test set, and a hallucination detector reaching **98.1% F1**.

Experience

Machine Learning Engineer & Dataset Team Lead | *Paradigm Networks, Inc.* **Jan 2023 – Present**

Dataset Team Lead — Data Curation & Data Engineering

Owned the data side of our in-house AI models, from raw data to training-ready datasets.

- Led a team in curating **two custom datasets of 10,000 data points each**, every record pairing a question, source context and answer with a quality label.
- Coordinated labelling work across the team, reviewed data quality, and resolved inconsistent or duplicate records.
- Built Python/Pandas pipelines to clean, validate, merge and split the data, combining in-house records with **64K open-source samples** into versioned training and test sets stored on AWS S3.
- These datasets directly enabled models that beat GPT-5 and Gemini on our internal benchmarks.

Python APIs & Backend Services

Production Python services behind the platform's AI features.

- Designed and built async **FastAPI REST APIs** for response verification, answer relevancy and faithfulness checks used by the main product.
- Built an **agent evaluation service** (FastAPI + MongoDB) that stores and scores every AI-agent run across 9 metrics, with results surfaced to users through the React web app.
- Integrated the services with a Go control server, a LangGraph agent platform and multiple LLM providers (OpenAI, Azure OpenAI, AWS Bedrock, GCP Vertex AI) with automatic provider fallback.
- Wrote CI test pipelines (pytest, mocked MongoDB) and released services through Docker, GitHub Actions, AWS CodeBuild/ECR and EKS.

LLM & Embedding-Model Solutions

Applied AI built into business workflows.

- Fine-tuned **IBM Granite (2B)** with LoRA for factuality detection (**91.7% accuracy**, ahead of GPT-5 and Claude 3 Sonnet) and hallucination detection (**98.1% F1**); serving both in production with vLLM.
- Built RAG verification using embeddings and chunk-level grounding checks (DeepEval **0.90 F1** vs. RAGAS 0.85).
- Built a LangGraph code-review agent that analyses pull-request diffs and publishes findings automatically, reducing manual review effort.

Teaching Assistant / Lab Engineer | *ITU Lahore — Mobile Application Development* **Jan 2024 – Jul 2024**

- Taught and mentored 100+ students, leading lab sessions and guiding final-year projects.

Full-Stack Developer Intern | *Clicky* **Jan 2022 – Oct 2022**

- Built and maintained full-stack web application features across frontend and backend.

Projects

Retrieval-Augmented Reasoning for Small LLMs | *Python, vLLM, FAISS, nomic/jina embeddings* Research project

- Built an embedding-based memory bank and a 600K-edge reasoning graph with FAISS vector search, feeding retrieved reasoning into a smaller Qwen3-1.7B model; raised accuracy from **75.0% to 85.4%**.

Visual Similarity Search | *PyTorch, ResNet-50, triplet loss* ML project

- Trained image embeddings with triplet loss for search-by-image across 101 categories; improved Recall@1 from 39.1% to 47.9%.

Technical Skills

Python & Backend:	Python, FastAPI, Django REST Framework, REST APIs, async programming, pytest, Go
Databases:	SQL, PostgreSQL, MongoDB, FAISS (vector search), data modelling
Data & Automation:	Pandas, NumPy, Excel data processing, ETL / data pipelines, data cleaning & labelling
Reporting:	Power BI, Tableau
AI / LLMs:	LLMs (GPT, Claude, Gemini, Granite, Qwen), embedding models, RAG, LangChain, LangGraph, fine-tuning (LoRA), Hugging Face, PyTorch, deep learning (Transformers, RNNs, LSTMs, GANs)
Cloud & DevOps:	AWS (S3, EKS, Bedrock), GCP, Azure, Docker, Kubernetes, GitHub Actions CI/CD
Web:	React, TypeScript, JavaScript, Next.js, HTML/CSS

Education

M.S. Data Science | *Information Technology University, Lahore* **2025 – 2027**

B.S. Computer Engineering | *Information Technology University, Lahore* **2019 – 2023**